Writing for teaching

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> **AI-Assisted Authorship**
>
> This chapter was written with the assistance of GitHub Copilot, which expanded on outlined ideas and draft notes provided by the author. The content represents the author's perspective and has been reviewed for accuracy, but the detailed prose was generated through AI assistance.

Lecture notes, slides, and problem sets teach rather than report. The general rules for scientific writing in the other chapters still apply. This chapter adds the rules that apply to teaching materials in particular.

Build the lecture from exercises and solutions

Active learning, in which students work through tasks themselves during class, improves exam performance over lecture alone (Freeman et al. 2014). The exercise-first structure is this guide's own convention for putting that result into practice. Pose each task as an exercise, give students time to try it, and then show the solution.

- Pose every data manipulation or analysis as an exercise, and show the code in the solution. Do not show the code first and explain it afterwards.
- Pose every calculation, derivation, or proof as an exercise, and put the worked steps in the solution.
- Ask the question that a definition or theorem answers before stating it. The solution answers informally, in plain words, and the formal `#def-` or `#thm-` div follows the solution div.
- Give each bottom-level prompt its own `#exr-` div and its own `#sol-` div, in order, rather than one exercise with parts (a) through (e). Any exercise with several parts is split, however the parts are marked: letters, numbers or bullets. Put the shared setup in the first exercise, and begin each later exercise by citing it, as in "Make the assumptions of the first exercise." Later exercises can use earlier results through cross-references.

- On slides, put a slide break between an exercise and its solution, so the prompt is on screen alone while students work.
- Give each solution the slug of its exercise, with `sol-` in place of `exr-`.
- Put a hint in a callout titled "Hint" inside the exercise div, not in italic text.

Example 0.1 (Asking before defining)**.**

Generalization is the ability of a model to perform well on new data.

```
::: {#exr-generalization}
What does it mean for a model to generalize?
:::

::: {#sol-generalization}
A model generalizes when it predicts well on data it was not fit to.
:::

::: {#def-generalization}
(the formal definition)
:::
```

The revised version lets students attempt the idea before they see the formal definition.

Build every unit from smaller atomic units

Make every unit of analysis as concise and minimal as possible. The units are mathematical expressions and divs of every kind: definitions, theorems, lemmas, corollaries, examples, remarks and proofs. When a unit grows complicated, do not write the complexity into it. Abstract the complexity out into smaller, atomic definitions, results and expressions, each with its own name and identifier, and glue the pieces together with hyperlinks.

This is the compositional approach of modular programming: a program is built from small, single-purpose modules that call one another, and a maintainer changes one module without reading the rest. The notes work the same way. A theorem links to the definitions it uses, a proof links to the lemmas it uses, and an expression is written with the named pieces it is built from.

- A display holds one idea. Name each meaningful sub-expression, give it a macro, and build the display from the names (see Making expressions concise).

- A div holds one term or one result. Move a second term, a side property or a long argument into its own div, and link to it from the place that needs it (see Give each defined term its own div).
- A proof states the lemmas it needs and links to them, instead of deriving them in the middle of the argument.
- A link replaces a repeated explanation. If a piece is already written, link to it instead of writing it a second time.

When a unit still looks long, ask what could be named and moved out.

> **Example 0.2** (Splitting one long definition)**.**
>
> One div defines the ridge estimator, defines its penalty, and proves that it shrinks the coefficients.
>
> Three divs: a definition of the penalty, a definition of the ridge estimator that links to the penalty, and a theorem on shrinkage that links to both.
>
> The version lets a later chapter link to the penalty alone, and each piece fits on a slide.

Give each defined term its own div

Defining terms clearly covers what a good definition contains. For teaching materials, four further rules keep definitions findable.

- Replace a bolded keyword in running prose with a `#def-` div. A bolded keyword usually marks an inline definition.
- Define one term per `#def-` div. A term defined inside another term's div, or two terms joined in one title ("Regression and classification"), has no identifier of its own, so nothing can link to it.
- Never nest one theorem-type div inside another. The rule covers `#def-`, `#thm-`, `#lem-`, `#cor-`, `#prp-`, `#cnj-`, `#exm-`, `#rem-`, `#exr-`, and `#sol-`. A special case, such as a linear evaluation function, gets its own div after the general case, and the two link to each other.
- Keep only the definition in a `#def-` div. Put the properties of the defined object, such as the symmetry and idempotence of a matrix, in a separate `#thm-`, `#lem-` or `#prp-` div after it, with a proof where one is known. Each abstract div then gets its own example, as the rule below requires.

Every alternative name for the same concept stays in the div of the term it names and is treated like the primary name: it appears in the div title and is bold where it first appears in the definition. Name synonyms and near-synonyms where the term is defined, and say which one the notes use. A student who meets one name here and another in a textbook cannot tell

whether the difference matters. For a near-synonym, name the seam: critical points include the stationary points and also the points where the derivative does not exist.

State the notation and its alternatives

When a symbol or convention has more than one common form, state the one the notes use and name the others a reader will meet elsewhere. Put the alternatives in the definition div or in a callout beside it. For example, some sources include 0 in the natural numbers and others exclude it, so write $\mathbb{N}_0$ or $\mathbb{N}^+$ to make the choice explicit.

Lead with the simplest correct form of a definition

Define a quantity in the form that needs the fewest symbols, and state the textbook form as a theorem that follows. For example, define variance as the mean squared deviation $\mathrm{E}\left[\mathrm{devn}(X)^2\right]$, where $\mathrm{devn}(X) = X - \mathrm{E}[X]$, and then prove that it equals $\mathrm{E}[X^2] - (\mathrm{E}[X])^2$.

Parsimony outranks convention when choosing a primary definition. The conventional form still appears in the notes, as a named result.

Reuse one written version of each result

When several courses or repositories have already written the same definition, theorem, or exercise, compare the versions on notation, decomposition, parsimony, and clarity. Use the best one, or merge them, and make it the single copy. Other places include or link to that copy instead of keeping a second wording that can drift.

Use real data and compute every number

- Where an example needs data, use a real dataset instead of invented round numbers. Name its original source, link to where it is published, and cite it.
- Show a dataset before analyzing it, with a printed look at its first rows and its structure.
- Show the data where it is first mentioned. Put a table of a few rows, or a plot of the variables, directly after the sentence that names the dataset, before any discussion of its features, the methods to apply to it, or the questions it raises. A reader who has seen the data can picture what the next slides describe, while a reader who has not must hold an abstract description in mind until the data appears. If the data are too large to show, show a sample or a summary figure. Link later mentions of the dataset back to this first display.

- Never type or paste output. The page shows what the code printed when the page was rendered, and each number in the prose comes from inline code. A number typed into prose goes stale when the data or the code changes.

Example 0.3 (Showing the data at first mention)**.**

> Slides 3 to 9 discuss the spam data's 57 predictors, its class imbalance and the loss we will minimize. Slide 10 shows a table of the first rows.
>
> Slide 3 introduces the spam data and shows a table of its first rows and a plot of one predictor by class. Slides 4 to 10 discuss the predictors, the class imbalance and the loss, and refer back to that table and plot.

The version lets every later slide refer to something the reader has already seen.

Store structured data in structured formats

This rule applies to all teaching materials, in every course and every teaching repository.

- Keep a list or table of like items, such as courses, readings, channels or glossary terms, in a CSV, YAML or JSON file with one record per item. The page reads the file and builds the table or list with a short function, instead of holding a hand-written Markdown table.
- A data file can be counted, filtered, sorted and reused by another page, and a new row is a one-line change that a reviewer reads as data.
- List the file under the page's `resources:` when the site publishes it.
- Convert an existing hand-written table in its own pull request, and render the page before and after to confirm that the output text is identical.
- A short list that no other page reads can stay as prose.

Show each idea applied and in a figure

- Give every definition, method and result at least one applied example that computes it on real data, so the reader sees what the idea does before or beside the algebra.
- Follow every abstract div directly with a concrete `#exm-` div. The rule covers each `#def-`, `#thm-`, `#lem-`, `#cor-` and similar div. The example of a related result does not exempt a definition: a theorem that uses a definition gets its own example, and so does the definition. The ai-config rule links here.
- Give every topic at least one figure that shows the idea, such as the data with a fitted curve, a distribution, or a diagram of the procedure. A page of definitions with no figure and no data is incomplete.
- Prerequisite notes follow the same rule as course lectures. A prerequisite page that is only definitions and algebra needs an applied example and a figure.

- Place each figure directly after the sentence that first refers to it. Do not refer to a figure, table, or result that appears later in the document, because the reader meets the reference before the thing it points to. This rule is for teaching materials, which are reading copies. A journal submission may collect its figures at the end, as Figures and tables describes.
- Make each figure follow the accessibility rules below.

Follow every example with an exercise and solution

Reading a worked example is passive. A reader who has just seen one method applied learns more by applying it to a new case than by reading a second example. So each example gets a practice exercise, with its solution, right after it.

- After every `#exm-` div, add an `#exr-` div that asks the reader to do the same thing for a different case, followed by its `#sol-` div. The exercise changes the data, the function or the question, so the solution is not a copy of the example.
- The full sequence for a new idea is therefore: the exercise that poses the question and its solution, the definition or theorem, an example, and a practice exercise with its solution.
- More generally, give readers an opportunity to think at every stage: a question they can answer before the next paragraph, a calculation to try, a prediction to make before the code runs.
- A counterexample (see Defining terms clearly) can stand in for the practice exercise when the exercise would be "decide whether this case fits the definition".
- Give the practice exercise its own slug and follow the numbering and slide-break rules above.

Example 0.4 (An example followed by a practice exercise)**.**

```
::: {#exm-sample-mean}
(the sample mean of one variable in a real dataset, computed by a code chunk)
:::

::: {#exr-sample-mean-new-variable}
Compute the sample mean of a different variable in the same dataset.
:::

::: {#sol-sample-mean-new-variable}
(the code chunk and its output)
:::
```

The example shows the method once. The exercise asks the reader to repeat it on a new

> variable before moving on, and the solution's numbers still come from running code (see Use real data and compute every number).

Write labs for both languages and any computer

These rules apply to all teaching materials, in every course and every teaching repository, not only to one course.

- Give labs and code exercises in both R and Python, side by side.
- Put each step's Python and R pair in its own tabset with `group="language"`. Start a new tabset whenever prose or a new step sits between code chunks, so the reader sees one step's code in one language at a time. The shared group makes one choice of language apply to every tabset on the page, as long as the tab headings use identical text in every tabset, such as always "R" and "Python".
- Write the R code with tidymodels, or show tidymodels and base R side by side. Do not give base R alone. ISLR-tidymodels-labs shows the style.
- When a lab follows a textbook's lab, use the book's exercises, data and fitting functions, in the book's order. For the ISL labs, fit Python models with `statsmodels`' `sm.OLS`, as ISLP_labs does. If an exercise departs from the book, say why in the lab.
- Never throw away an exercise you replace. Either keep both exercises, or keep the replaced one as an outtake: leave its file in place and stop including it.
- Tell students where to get the data.
- Point students only to places they can open. Students read a course's published site, not its repository, which is often private. So never send them to a file in the repository, to a `github.com` link into a private repository, or to "the copy in this repository". Link a data file from the published site, from a public repository or package, or from the data's original public source.
- Do not assume that students own a laptop. Assume only that they can use some computer, so setup steps must work on a shared or hosted machine.

Present R results with modern packages

This rule applies to all R code in teaching materials, in every course and every teaching repository.

- Even when you fit a model with a base R function such as `lm()` or `glm()`, show its results with modern packages, not with base R output.
- Show coefficient tables with parameters (`parameters() |> print_md()`) or gtsummary (`tbl_regression()`), not with `summary()` or `coef(summary())` printouts.
- Draw figures with ggplot2, not with base graphics such as `plot()`, `lines()` and `abline()`.

- Draw model plots with packages built on ggplot2: sjPlot or ggeffects for predictions, and performance's `check_model()` for diagnostic plots.
- Show model fit statistics with performance's `model_performance()` or broom's `glance()`.
- Ezra's rme notes show the style.

Manipulate data with dplyr, not base R

This rule applies to all R code in teaching materials, in every course and every teaching repository.

- Select columns with dplyr's `select()`, not with `df[, c("a", "b")]` or `df[c("a", "b")]`.
- Filter rows with `filter()`, not with `df[df$x > 0, ]` or `subset()`.
- Add or change columns with `mutate()`, not with `df$new <- ...`.
- Drop rows with missing values with tidyr's `drop_na()`, not with `df[complete.cases(df), ]`.
- Stack data frames with `bind_rows()`, not with `rbind()`.
- Build data frames with tibble's `tibble()`, not with `data.frame()`.
- Chain steps with the native pipe `|>`.
- Do not nest function calls. Write `tibble(x, y) |> pander()`, not `pander(tibble(x, y))`, and `sizes |> lapply(f) |> bind_rows()`, not `bind_rows(lapply(sizes, f))`. Give a call that would sit inside an argument, such as `data = filter(df, train)` in a model call, a name on its own line first.
- Reading one value out of a data frame for inline code is fine, as is indexing a matrix or a vector. The rule is about selecting, filtering and building data frames.

Include code from files instead of copying it

This rule applies to code that appears in more than one place in teaching materials.

- When the same code appears in several pages, or in a page and a script, keep one source file and include it where it is shown. A copy in each place drifts when one copy is fixed and the others are not (see Reuse one written version of each result).
- Code that the page runs should be an executed chunk, as above. For code that is displayed but not executed on the page, such as a standalone script or a configuration file, Quarto's include-code-files extension fills a code block from a file. Add `filters: [include-code-files]` to the front matter, then write a code block with an `include` attribute, such as ```` ```{.python include="script.py"} ````.
- The extension also takes `start-line` and `end-line`, to show part of a file. Prefer a range that names the lines you want over a copy of those lines.
- The extension is a new dependency of the site: install it with `quarto add quarto-ext/include-code-files`, and commit the `_extensions` directory it creates.

Comment every code chunk to say what and why

This rule applies to all reader-facing code in teaching materials, in every course and every teaching repository, in every language.

- Put comments in each code chunk that explain both what the code is doing and why it does it.
- Leave out a comment that would give away the solution in an exercise prompt. Add the comments to the solution's code instead.
- A comment that only restates the code, such as `# add 1 to x` above `x <- x + 1`, gives the *what* without the *why*. Say what the step accomplishes and why the analysis needs it, for example `# center age so the intercept is the mean outcome at the average age`.
- Comment each step that a student could not reconstruct from its name, such as a transformation, a modeling choice, a tuning value or a random seed.

Set `code-fold` by what the code does and whether the prose mentions it

This rule applies to every executed code chunk in teaching materials, in every course and every teaching repository.

Set `#| code-fold:` on each chunk by two questions: does the prose mention the code, and does the code produce output outside the chunk, such as a figure, a table, or variables that inline expressions use?

- `#| code-fold: show` when the prose mentions the code and the code produces output outside the chunk: a figure, a table, or variables that inline expressions use. The code starts open, and a reader who wants to focus on the output can hide it.
- `#| code-fold: true` when the prose does not mention the code. This holds whether the code produces output outside the chunk or no output at all. The code starts hidden, and a reader who wants it can open it.
- `#| code-fold: false` when the prose mentions the code and the code produces no output, or only output that the chunk renders inside itself. The code and its output stay together, with no toggle.

For example, a chunk that fits a model the text walks through and prints its coefficients is `false`, a chunk that draws a figure the text discusses is `show`, and a chunk that loads a package or builds an object that later chunks use is `true`.

Introduce every function in a programming primer

This rule applies to all teaching materials that run code, in every course and every teaching repository.

- Give the course a programming primer that introduces each package function the course's chapters call. A student meets the function there first, in a short note, a worked example and an exercise, and then sees it used in the chapters.
- Keep the primer complete by computing the list of functions, not by remembering it. Scan the code chunks of the chapters for the functions they call, and compare the list with the functions the primer mentions. Run the scan when a chapter adds a function, and again before a term starts.
- Add the missing functions in separate pull requests, one per package. A pull request that adds a function to a chapter adds it to the primer too, or links the primer entry that already introduces it.
- Functions that a chapter defines for itself are not package functions. A chapter introduces those where it defines them.

Make figures accessible

- Wrap each figure, including an interactive one, in a `#fig-` div with a caption.
- Give every plot alt text. For Observable Plot, set `ariaLabel`.
- Show the code that makes each figure, and let readers fold it away. Code folding does nothing for a cell that does not echo its code, so do not set `echo: false` on a cell you want students to read.
- Do not cross-reference an interactive figure from prose, because the PDF format renders no figure for it, so the reference has no target there (observed when rendering lecture notes to PDF).

Structure slides for the reader

- Check the rendered slides, not only the web page. Slides need the same boxes and colors on theorem-type and callout divs that the web page has, so a reader can see where a definition ends and the commentary begins.
- Put a list of three or more phrases in a bullet list, as Put lists in bullet points describes.
- Put each named example in an `#exm-` div.
- Put on a slide only what you want the audience to look at. Put an example in an `#exm-` div and short commentary in a `#rem-` div, because a `::: notes` div is plain text without a box on the web page.

- Put a long paragraph of commentary in a remark that is also speaker notes: `::: {#rem-name .remark .notes}`. Reveal.js shows it only in speaker view, so the audience does not have to read it, and the web page shows it as a boxed remark. Readers who did not attend the lecture use the web page, so nothing is lost, and nothing is said twice. Give a long paragraph a separate short `#rem-` div only when the slide needs a takeaway to look at.
- Add structure only where content has a shape. Do not put a heading over a single item, or over the same topic as the heading just before it.
- Keep body text off section and title slides. Reveal.js centers those slides vertically and does not scroll them, so long text is clipped at the top and bottom of the screen (observed in lecture slides rendered with Quarto). Put a slide break after the heading and move the text onto content slides.

Give each concept in a note sheet its own row

A note sheet is a reference table of formulas with brief labels and no prose. This is one case of building every unit from smaller atomic units. Give each concept its own row. A row holds one formula and one short label.

- Never put two definitions in one row.
- Never merge rows to save space. Find the room another way, such as a smaller table font or a second table beside the first.
- Never put notation or assumptions inside a label. Give each assumption its own row, and state the notation once in its own row.
- Write each formula in its most compact form, built from pieces that have their own rows (see Making expressions concise).

The same rule applies to every other reference table, such as a table of notation.

Example 0.5 (Splitting a merged row)**.**

Row: `Dot product, sample mean`, with both formulas in one cell.

Rows: `Dot product` with its formula, then `Sample mean` with its formula.

The row makes the reader find where one concept ends and the next begins. The rows let the reader find one formula by one label.

Talk about the ideas, not the notes

When a term is defined on another page or in another site, state the idea in the sentence and link the term. Do not write about where the definition lives.

- Write "the bias of an estimator", linked to its definition, not "the statistics notes define the bias of an estimator".
- Do not write "see the probability notes" or "as in our notes". The link already says where the term lives.
- Name a remark by the idea it holds, such as "Bias of a prediction", not "Bias of a prediction in the statistics notes".

Students read concepts, not the structure of the site. A sentence about which notes define a term adds no content, and it goes stale when the definition moves.

Credit other courses without summarizing them

- Write your own version of another course's material, and credit the source in a collapsed `.callout-note` titled "Source", not in running prose. Adapting another course's material gives the details, including when a `::: notes` div is a safe alternative.
- Quote the primary source directly, and check the wording and page against the original. Quote the original explains why.
- Title a section by its topic, not by its author or course. Put the content first explains why.
- Where the book is the subject rather than its authors, name the book by its title, because a bare citation key renders as the author list.

Rules that apply to all writing

These rules from the other chapters apply to teaching materials as to any other writing:

- start a sentence with its subject;
- name the referent instead of a bare demonstrative;
- say which thing;
- state the literal claim instead of an idiom;
- avoid clichés.

Freeman, Scott, Sarah L. Eddy, Miles McDonough, et al. 2014. "Active Learning Increases Student Performance in Science, Engineering, and Mathematics." *Proceedings of the National Academy of Sciences* 111 (23): 8410–15. https://doi.org/10.1073/pnas.1319030111.